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The PhenoMeNals framework for grapevine yield and quality prediction

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ABSTRACT

Timely prediction of grapevine yield and quality is a major scientific challenge with direct implications for vineyard management, enological planning, and winery logistics. Most existing approaches remain limited to specific sites or varieties and do not account for the two-year grapevine cycle, thereby overlooking the temporal dynamics underlying reproductive physiology. Here, we introduce PhenoMeNals, an open-source framework integrating phenology simulation with process-based ecophysiological functions to forecast grapevine yield and quality, explicitly considering carry-over effects. The coupled dormancy-forcing phenology model, calibrated with BBCH observations (RMSE 6-9 days), is the foundation of the workflow, and provides the temporal structure spanning the two-season grapevine cycle. Weather cues are sigmoid normalized using long-term percentile distributions, weighted by their correlation strength with target traits (Pearson correlation), and cumulatively summed along the phenological timeline to derive memory signals (i.e., PhenoMeNals). Multicollinearity is addressed through pairwise filtering and stepwise feature selection, and model robustness is ensured using leave-one-out cross-validation. The framework was evaluated using multi-site, multi-variety datasets encompassing yield, yield components, and quality traits. Predictive performance was comparable or superior to published models (R^2 consistently above 0.8), with gaining accuracy as the seasons progressed. Beyond prediction, PhenoMeNals provides phase-specific metrics that enable interpretation of weather influence on reproductive development. By combining physiological realism, statistical robustness, and open-source release as R package, PhenoMeNals fosters reproducibility, cross-site comparison, and user-driven adaptation.

1. Introduction

The early forecasting of fruit yield and quality has become a distinct discipline in agrometeorological research (He et al., 2022; Magro et al., 2023), with viticulture representing one of the most studied crops (Victorino et al., 2020; Moreira et al., 2025). Besides its scientific relevance, grapevine yield prediction also holds significant operational value, as timely estimations support harvest scheduling, resource allocation, labor and machinery optimization, logistics, and strategic decisions related to market positioning and pricing (Taylor et al., 2019; Palacios et al., 2022).

A wide range of methodologies has been developed for this purpose, spanning from direct methods based on manual or digital fruit detection and counting, to modeling strategies based on ecophysiological relationships linking grapevine yield and quality to management practices, weather conditions, and remotely sensed vegetation indices (Laurent et al., 2021; He et al., 2022). Increasingly, hybrid grapevine yield prediction models employing empirical and process-based logic are being developed and applied across spatial and temporal scales (Santos et al., 2012; Massano et al., 2023). Regardless of the specific approach, explicitly accounting for phenology is required to establish biologically meaningful relationships, as the timing of reproductive phases shapes the plant sensitivity to weather cues (Cameron et al., 2022; Delelee et al., 2025). This consideration is especially critical in perennial crops, where fruits formation generally spans two growing seasons and is influenced by biochemical and molecular regulatory mechanisms interacting with weather conditions in specific phenophases (Li-Mallet et al., 2015).

In grapevine, the interplay between plant ecophysiology and weather conditions is reflected into the determination of yield components, i.e., bunch number, flower and berry number, and berry mass, which are shaped by phase-specific responses to temperature, light, water availability, and nutrient status (Guilpart et al., 2014). The formation of these components starts from the initiation of primordia in latent buds during the preceding season and unfolds in the harvest year through flowering and fruit ripening (Vasconcelos et al., 2009). Yield

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quality traits, such as sugar accumulation, acidity, pH and aromatic profiles, are also influenced by the timing and intensity of exposure to weather conditions during key phenophases, often through complex metabolic pathways (Duchêne, 2016; Faralli et al., 2024).

Although extensive research has been conducted on how climatic factors affect grapevine yield and quality, the resulting knowledge remains poorly harmonized. Most studies focus on specific cultivars, regions, or time frames, with limited generalizability. As a result, experimental evidence is difficult to integrate across diverse viticultural contexts, and predictive models are rarely tested on datasets spanning multiple environments and genotypes, hindering their practical applicability for vineyard management (Barriguinha et al., 2021). In addition, most forecasting models rely on statistical correlations defined over fixed time windows or specific phenophases, neglecting carry-over effects and cross-seasonal interactions inherent to the grapevine reproductive cycle (e.g., González-Fernández et al., 2020; Ghiglieno et al., 2023). The absence of a phenology-aware approach accounting for the carry-over effects of weather cues hinders systematic and reproducible analyses across varieties and growing conditions. This limitation is even more critical given that climate change has already altered known grapevine phenological patterns, thereby disrupting the timing and magnitude of associated physiological responses (Alikadic et al., 2019).

In response to these needs, we developed PhenoMeNals (Phenology Memory sigNals), a predictive framework that combines phenology modeling with cumulative memory signals derived from ecophysiological response functions (EcoPhysFs). The aim of PhenoMeNals is to capture the progressive influence of weather conditions on yield and quality traits, enabling biologically grounded relationships and comparisons across grapevine cultivars and environments. We tested the framework on a wide range of site × variety combinations to evaluate its robustness, reproducibility, and applicability.

2. Materials and methods

2.1. Overview of the PhenoMeNals framework and validation dataset

The PhenoMeNals framework reconstructs the cumulative influence of weather cues on yield and quality across the two-year grapevine reproductive cycle by integrating phenology modeling with process-based logic (Fig. 1). To assess its validity and performance, we

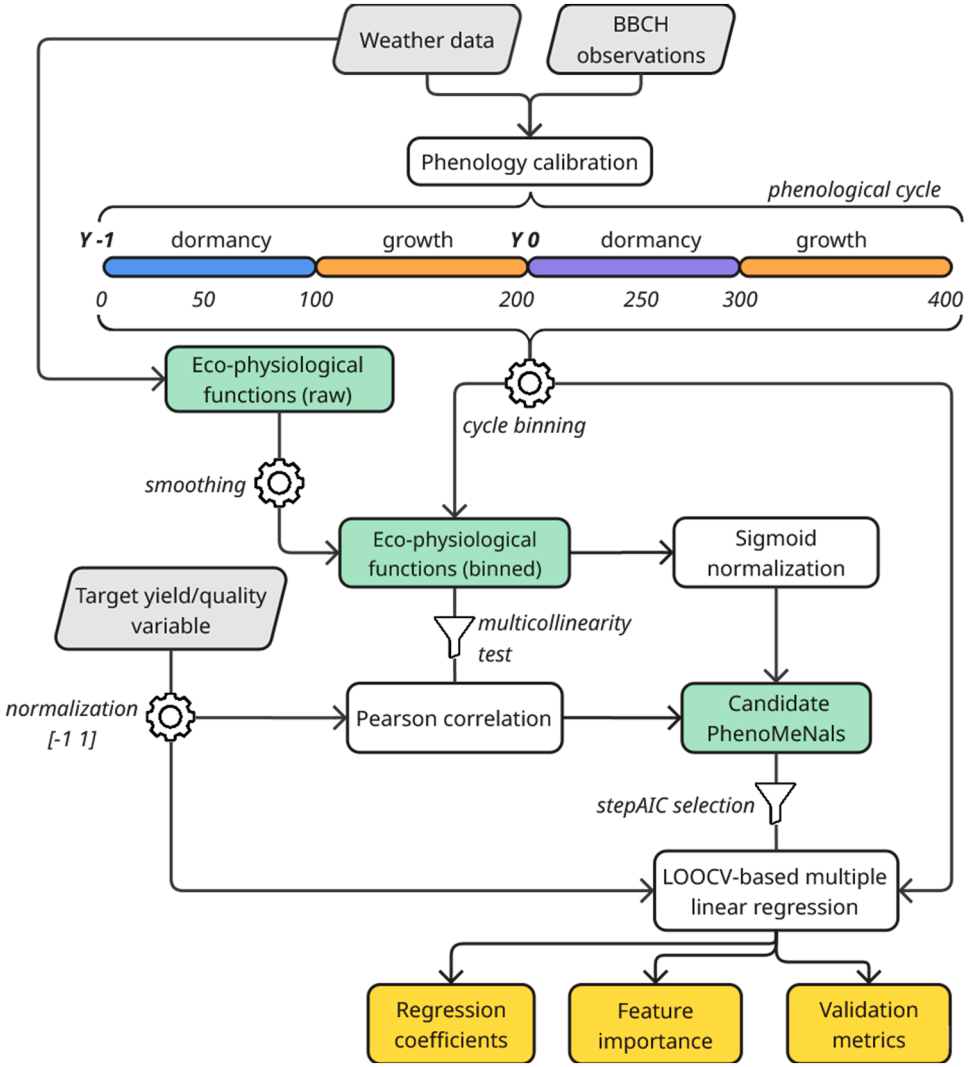


Fig. 1. Outline of the PhenoMeNals framework. The worflow integrates weather data and BBCH-scale observations to calibrate a phenological model, reproducing the grapevine cycle from dormancy in the previous season (Y-1) to harvest in the current season (Y0). Along this timeline, eight ecophysiological functions are computed and aggregated within phenological bins (0-400). Each function is evaluated for its correlation with the target trait and normalized via sigmoid transformation. The resulting cumulative memory signals (candidate PhenoMeNals) serve as predictors for Leave One Out Cross Validation (LOOCV) via multiple linear regression. Gears indicate internal operations (smoothing, binning, normalization); funnels indicate steps of variable evaluation or selection (multicollinearity test and stepAIC feature selection; yellow boxes are final outputs, green boxes are intermediate outputs, and white boxes are statistical methods.

compiled reference data from scientific literature, selecting studies that reported BBCH phenological observations together with target traits such as yield, yield components (e.g., bunch number, berry weight), and quality attributes (e.g., sugar content, Brix).

The validation dataset included six vineyard sites across Europe (Italy, Germany, Spain, Luxembourg), Oceania (Australia), and North America (USA), with time series ranging from 12 to 47 years depending on the site (Table 1).

Table 1
Test sites for the PhenoMeNals framework, including country, available years and observation period, target traits, reported BBCH phenological stages, grapevine varieties (abbreviations explained in the footnotes), source of weather data, and reference to the original study. Site coordinates are included for reference (*approximate when not explicitly reported in the original publication).

Site	Years	Target trait	BBCH	Variety	Weather	Ref
COL* *lat 46.00° N lon 13.44° E*	2006 -2024 (18)	YIE, BUN, BER, SUG	8, 65, 81, 89	CAB, MER, CAR, PIC, PIG, REF, RIB, SAU, SCH, TOC, VER	ARPA FVG	a
NAP *lat 35.43° N lon -119.05° E*	1994 -2018 (25)	SUG	-	CAB, MER, SAU, CHA	NASA-POWER	b
OYS* *lat -41.30° S lon 174.11° E*	2005 -2019 (15)	YIE	8, 65, 81, 89	SAU	AGBM	c
PEN *lat 41.53° N lon 1.77° E*	1998 -2012 (15)	YIE	8, 65, 81, 89	CHA	E-OBS	d
LUX *lat 49.54° N lon 6.35° E*	1993 -2015 (23)	YIE	8, 63, 68, 81, 89	MUL, RIE	E-OBS	e
WUR *lat 49.48° N lon 9.56° E*	1962 -2010 (47)	YIE, SUG	8, 65, 81, 89	RIE	E-OBS	f

Abbreviations:
Sites: COL = Colli Orientali, Italy ; NAP = Napa Valley, United States of America; OYS = Oyster Bay, Australia; PEN = Penedes, Spain; LUX = Remich, Luxembourg ; WUR = Wurzburg, Germany.
Target trait: YIE = yield; BUN = bunch number; BER = berry weight; SUG = sugar content; see NSupplementary Table S2 for units, which differ across sites.
Varieties: CAB = Cabernet Sauvignon; MER = Merlot; CAR = Carmenere; PIC = Picolit; PIG = Pignolo; PIB = Pinot Blanc; REF = Refosco; RIB = Ribolla; TOC = Tocai; SAU = Sauvignon Blanc; SCH = Schioppettino; VER = Verduzzo; CHA = Chardonnay; MUL = Muller Thurgau; RIE = Riesling
Reference: a = Consorzio Tutela Vini Friuli Colli Orientali e Ramandolo (2006-2024); b = Bai et al. (2022); c = Parker et al. (2020); d = Leolini et al. (2018); e=Molitor and Keller (2016); f=Bock et al. (2013)
BBCH phase: 08 = Bud burst: green shoot tips clearly visible; 63 = Early flowering: 30% of flowerhoods fallen; 65 = Full flowering: 50% of flowerhoods fallen; 68 = 80% of flowerhoods fallen; 81 = Beginning of ripening: berries begin to develop variety-specific colour; 89 = Berries ripe for harvest
Weather source: ARPAFVG = Friuli Venezia Giulia regional agency for environmental protection (https://www.arpa.fvg.it/); AGBM = Australian Government Bureau of Meteorology (http://www.bom.gov.au/climate/data/); NASA-POWER = NASA Prediction of Worldwide Energy Resources (https://power.larc.nasa.gov/); E-OBS = Ensemble daily gridded observational dataset in Europe (https://surfobs.climate.copernicus.eu/dataaccess/access_eobs.php)

Grapevine yield was the most consistently available target trait, present at nearly all sites, while additional traits included yield components (bunch number, berry number, berry weight) and fruit quality variables such as sugar content (g l^{-1}) and soluble solids (°Brix). Phenological development was recorded at key BBCH stages including budburst (8), flowering (63, 65, 68), veraison (81), and harvest (89). For sites lacking direct BBCH observations (Napa Valley, US), the framework was tested using phenology parameters calibrated for the same variety at other locations. Daily weather data were obtained from institutional sources (e.g., ARPAFVG, Australian Bureau of Meteorology) and global gridded datasets (e.g., NASA-POWER, E-OBS). Variables include daily maximum and minimum temperature (°C), precipitation (mm), solar radiation (MJ m^{-2} d^{-1}), humidity, and wind speed (m s^{-1}, when available). The data were retrieved at daily resolution and hourly weather profiles were internally generated from these inputs (Supplementary S1). When variables were only reported graphically, data were manually digitized using WebPlotDigitizer v.3.4. All reference and weather data used in this paper are freely released as electronic tables in Excel format (Supplementary Table S2).

2.2. Forecasting methodology

2.2.1. Phenology model calibration

The workflow begins with the simulation of grapevine phenology using a sequential chill-forcing model previously validated by Bregaglio et al. (2025) across diverse grapevine varieties in Italy. The model divides the dormancy season into two phases: a period (endodormancy), during which bud growth is physiologically inhibited, and a quiescent period (ecodormancy), where dormancy is maintained by unfavorable weather conditions (Lundell et al., 2020). Progression through dormancy is modeled by accumulating chill days and anti-chill days using the single triangle method based on fixed temperature thresholds (see Cesaraccio et al., 2004 for the original formulation). The rest phase ends once a predefined chilling requirement is fulfilled, after which anti-chill accumulation leads to budbreak. Following dormancy release, phenological development is simulated via growing degree day (GDD) accumulation, calculated from a nonlinear hourly response function defined by minimum, optimum, and maximum temperatures (eq. 1, Yan and Hunt, 1999). Progression through phenophases (BBCH scale) is determined by variety-specific GDD requirements (Bregaglio et al., 2022). Model parameters, including chilling and forcing thresholds, are estimated using a simplex algorithm (Huang et al., 1998) that minimizes the Root Mean Square Error (RMSE, days) between observed and simulated BBCH. To integrate weather cues with phenological development, the grapevine reproductive cycle is mapped onto a standardized continuous timeline spanning two consecutive growing seasons, which is subsequently discretized into fixed binned intervals (from 0 to 400). The preceding season (Y-1) spans bins 0–200, while the harvest season (Y0) spans bins 201–400. Dormancy in Y-1 is represented by bins 0–100, including chill accumulation (0–50) and anti-chill (51–100), followed by the Y-1 growing season (bins 101–200). This is followed by dormancy in Y0 (bins 201–300), including chill accumulation (201–250) and anti-chill (251–300), and finally by the Y0 growing season (bins 301–400).

2.2.2. Ecophysiological functions

Along with the above-mentioned standardized phenological timeline, eight EcoPhysFs driven by weather conditions and biologically relevant parameters are computed (Fig. 2). Parameter values were derived from previously published studies, with the corresponding sources provided in Supplementary Table S3.

These EcoPhysFs include: i) temperature suitability for growth (TempF, Fig. 2a); ii) cold (ColdF, Fig. 2b) and iii) heat (HeatF, Fig. 2c) stress; iv) radiation-driven limitation (LightF, Fig. 2d); v) vapor pressure deficit limitation (VPDeF, Fig. 2e); vi) evaporative demand (DroughtF, Fig. 2f); vii) wind-induced stress (WindF, Fig. 2g); and viii) disease

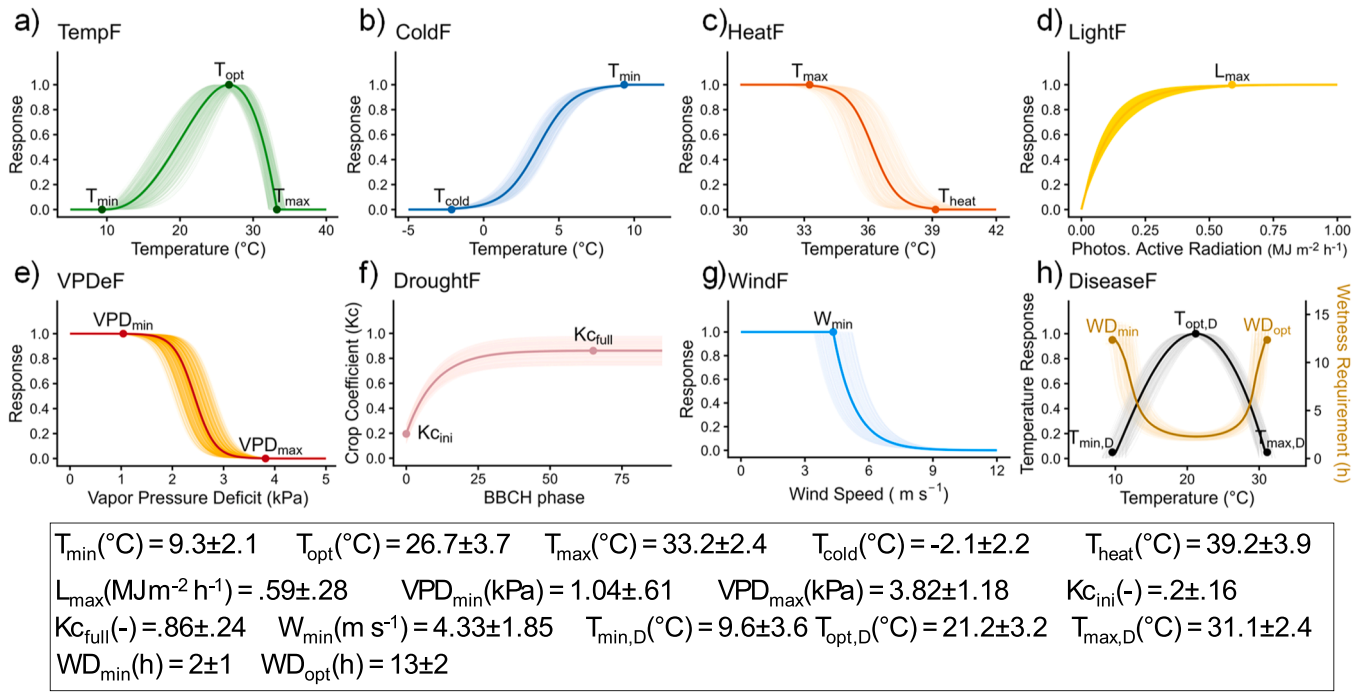


Fig. 2. Ecophysiological functions used in the PhenoMeNals framework. Each function is parameterized from literature search (white box). Shaded areas represent uncertainty across parameter estimates. (a) Temperature suitability for growth (*TempF*), driven by minimum (T_{min}), optimum (T_{opt}), and maximum (T_{max}) temperature; (b) Cold stress response (*ColdF*), with sigmoidal decline below a threshold (T_{cold}); (c) Heat stress response (*HeatF*), with sigmoidal decline above a threshold (T_{heat}); (d) Radiation limitation (*LightF*), with exponential saturation above a threshold (L_{max}); (e) Vapor pressure deficit limitation (*VPDeF*), with sigmoidal decay between VPD_{min} and VPD_{max}; (f) Crop coefficient dynamic along BBCH stages from initial (Kc_{ini}) to full canopy (Kc_{full}) used in *DroughtF* (see the main text); (g) Wind-induced stress (*WindF*), with exponential decrease beyond a threshold (W_{min}); (h) Disease favorability (*DiseaseF*) based on temperature response (the same function as in *TempF*) and wetness duration requirements.

suitability for pathogen infection, using *Plasmopara viticola* (causal agent of grapevine downy mildew) as the reference organism (DiseaseF, Fig. 2h). Each EcoPhysF is computed at an hourly time step, aggregated to daily values, and normalized from 0 (inhibitory conditions) to 1 (optimal conditions). TempF (eq. 1) is a nonlinear response function reproducing the physiological relationship between air temperature (T,°C) and plant growth rate (Yan and Hunt, 1999). It is driven by three cardinal temperatures as parameters (T_{min}, T_{opt}, T_{max};°C), with the peak response at T_{opt}, and zero below T_{min} or above T_{max} (Fig. 2a).

$$TempF = \begin{cases} 0 & T \le T_{min} \text{ or } T \ge T_{max} \\ \left(\frac{T_{max}-T}{T_{max}-T_{opt}}\right) \times \left(\frac{T-T_{min}}{T_{opt}-T_{min}}\right)^{\left(\frac{T_{opt}-T_{min}}{T_{max}-T_{opt}}\right)} \end{cases} \quad (1)$$

ColdF and HeatF represent the effect of temperature extremes on physiological response. While previously implemented as linear functions (Bregaglio et al., 2024), they have been formulated using sigmoidal response curves to capture the nonlinear effects of temperature beyond the optimal range on plant physiological processes. ColdF (eq. 2) reproduces the sensitivity to low temperatures, with response decreasing as temperature falls below T_{min} and reaching zero at the critical cold threshold T_{cold} (°C):

$$ColdF = \begin{cases} 1 & T > T_{min} \\ 0 & T < T_{cold} \\ \frac{1}{1+\exp(-10/|T_{min}-T_{cold}| \times (T-(T_{min}+T_{cold})/2))} \end{cases} \quad (2)$$

HeatF (Eq. 3), captures the decline in physiological activity due to high temperatures, decreasing beyond T_{max} and approaching zero at the critical heat temperature T_{heat} (°C).

$$HeatF = \begin{cases} 1 & T < T_{max} \\ 0 & T > T_{heat} \\ \frac{1}{1+\exp(10/(T_{max}-T_{heat}) \times (T-(T_{max}+T_{heat})/2))} \end{cases} \quad (3)$$

Both the light (LightF) and vapor pressure deficit (VPDeF) response functions are derived from the Jarvis-type stomatal conductance model (Jarvis et al., 1976) and are applied consistently throughout the crop cycle. The LightF function (Eq. 4) models the effect of photosynthetically active radiation (PAR, MJ m^{-2} h^{-1}) on stomatal conductance.

$$LightF = 1 - \exp\left(-k_{light} \times \frac{PAR}{L_{max}}\right) \quad (4)$$

It follows an asymptotic saturation curve to simulate the limitation of photosynthetic activity under low irradiance and the progressive saturation as radiation increases toward $Light_{max}$ (Raab et al., 2015). The shape parameter k_{light} (unitless, default: 5) regulates the steepness of the response, with higher values leading to a more rapid transition toward light-saturated conditions. The VPDeF function (Eq. 5) represents the limiting effect of atmospheric evaporative demand on stomatal conductance.

$$VPDeF = \begin{cases} 1 & VPD < VPDmin \\ \frac{1}{1+\exp(k_{VPD} \times (VPD-(VPD_{max}+VPD_{min})/2))} \end{cases} \quad (5)$$

Unlike linear formulations (e.g., Manzini et al., 2024), VPDeF is expressed in a nonlinear form to capture the gradual onset and intensification of stress with increasing vapor pressure deficit (VPD). No limitation is applied below a minimum threshold (VPD_{min}), while above this point, the function declines sigmoidally, with the steepness

governed by the shape parameter k_{VPD} (unitless, default: 5). DroughtF (Eq. 6) is an aridity indicator defined as the normalized difference between crop transpiration (ETc, mm) and precipitation (Prec, mm) (Confalonieri et al., 2012).

$$DroughtF = \left[\left(\frac{ET_c - Prec}{ET_c + Prec}\right) + 1\right] \Big/ 2 \quad (6)$$

The indicator is computed using a rolling average over a user-defined period (default: 15 days) to smooth short-term variability and reflect cumulative water balance dynamics. Crop transpiration is estimated from reference evapotranspiration (ET_o, mm d^{-1}), scaled by a phenology-dependent crop coefficient (Kc, unitless, Eq. 7).

$$Kc = \begin{cases} Kc_{\max} & BBCH > 65 \\ Kc_{ini} + (Kc_{full} - Kc_{ini}) \times \dfrac{1 - \exp(-0.1 \times BBCH)}{1 - \exp(-0.1 \times 65)} \end{cases} \quad (7)$$

The Kc remains constant at Kc_{ini} during dormancy and begins an exponential increase at budbreak, progressively approaching Kc_{full} by full flowering (BBCH 65) (Williams et al., 2022). From that phenological phase onward, it stays constant through the end of the crop cycle (BBCH 89, corresponding to maturity). The WindF response function (Eq. 8) accounts for the potential stress induced by excessive wind speed on stomatal closure, increased transpiration, and mechanical stress (Freeman et al., 1982; Alonso et al., 2024).

$$WindF = \begin{cases} 1 & W < W_{\min} \\ \exp(-k_{wind} \times (W - W_{\min})) \end{cases} \quad (8)$$

This function remains equal to 1 when wind speed (W, m s^{-1}) is below a defined threshold (W_{min}, m s^{-1}), beyond which it decays exponentially with the shape parameter k_{wind} (unitless, default: 1) regulating the sensitivity to high wind speed. As with the other functions, it is applied continuously across the crop cycle. The DiseaseF response function (eq. 9; Magarey et al., 2005) evaluates the weather suitability to *P. viticola* infection by combining temperature and leaf wetness duration.

$$DiseaseF = \begin{cases} 1 & W > W_{opt} \\ 0 & W < W_{min} \\ \dfrac{W_{min}}{f(T)} \Big/ W_{opt} \end{cases} \quad (9)$$

Temperature response is modeled using the beta-type function in eq. 1 parameterized for the pathogen thermal requirements. This response is then used to scale the minimum required wetness duration (WD_{min}) based on how favorable the current temperature (T) is during a leaf wetness period (Eq. 9). As temperature approaches the pathogen's optimum (T_{opt}), the required wetness duration decreases (Fig. 2h). If the scaled wetness value exceeds a maximum threshold (WD_{opt}), conditions are considered fully favorable (1). The function is not computed during wetness periods when temperatures fall outside the defined physiological range. The eight EcoPhysFs are derived from established modeling approaches, and their default parameterization is based on an extensive literature review (Fig. 2 and Supplementary S3). Users can customize both biologically relevant parameters and empirical coefficients to adapt the functions to specific crops, environments, or experimental objectives.

2.2.3. Phenology-based binning, multicollinearity and correlation analysis

Each EcoPhysF is first smoothed using a rolling mean over a user-defined time window to reduce short-term noise (default: 3 days) and then discretized into bins along the phenological timeline (bins 0-400, default: 1 bin interval). This phenology-based binning ensures that weather cues are aligned with biologically meaningful stages of the reproductive cycle, rather than fixed calendar dates. To address potential multicollinearity among EcoPhysFs, pairwise Pearson correlations are calculated between their binned values. Values exceeding a user-defined threshold (default: 0.9) are screened, retaining those with the lowest overall correlation with the remaining EcoPhysFs. For each phenological bin, the Pearson correlation coefficient and associated p-value are computed between the binned EcoPhysF and the target traits (e.g., yield, sugar content) across years. In practice, for each bin the EcoPhysF values from all available years are correlated with the corresponding trait values observed in those same years. To ensure comparability across traits and sites, trait values are min–max normalized to the [–1, 1] range within each site × variety combination. This process generates a continuous trajectory of correlation across the phenological bins for each EcoPhysF, which is then used to identify the developmental stages where each variable contributes most strongly to trait variability.

2.2.4. Sigmoid scaling, weighting and cumulative signal construction

This step transforms the EcoPhysF trajectories into the phenology-based memory signal (i.e., the PhenoMeNal) by (i) scaling yearly EcoPhysF values relative to their interannual distribution within each phenological bin, (ii) weighting them according to their statistical association with the target trait, and (iii) cumulatively integrating these contributions along the phenological timeline. Following binning and correlation analysis, EcoPhysFs are normalized within each phenological bin to quantify their relative expression across years. For each EcoPhysF and phenological bin, the distribution of values across all available years is used to compute the 10th, 50th, and 90th percentiles. These statistics are therefore derived from the interannual distribution of the binned EcoPhysF values rather than from the original daily weather data. Thus, if *n* years are available, the percentiles for each bin are computed from the *n* EcoPhysF values associated with that bin. A sigmoid transformation is then applied to map EcoPhysF values onto a bounded [0,1] scale (s_i, Eq. 10), reflecting their relative position within the interannual distribution.

$$scaled_i = \frac{1}{1 + \exp\left(-10 \times \dfrac{f_i - \tilde{f}}{(f_{90} - f_{10})/2}\right)} \quad (10)$$

where f_i is the value of the EcoPhysF in year i, $\tilde{f}$ is the bin-specific median, and f_{10}, and f_{90} are the 10th and 90th percentiles. Each scaled EcoPhysF is then weighted according to its association strength with the target trait. Specifically, the scaled value is multiplied by the Pearson correlation coefficient and adjusted by the significance of the relationship (Eq. 11), producing the corresponding EcoPhysF weighted value.

$$weighted_i = scaled_i \times r \times (1 - p) \quad (11)$$

where $weighted_i$ (0-1) is the weighted value of the EcoPhysF in year i, r is the Pearson correlation coefficient between the EcoPhysF and the target trait, and p its associated p-value. In this formulation, the Pearson r captures the strength and direction of the EcoPhysF-trait association, while the term (1-p) acts as an attenuation factor that reduces the contribution of relationships with weaker statistical support. This weighting step ensures that phenological periods showing strong and statistically supported associations with the target trait contribute more prominently to the cumulative phenology-based memory signal (i.e., the PhenoMeNal), while deviations occurring in weakly related periods have limited influence. Finally, the PhenoMeNal is obtained by summing the weighted EcoPhysF values along the phenological timeline (Eq. 12).

$$PhenoMeNal_{i,j} = \sum_{bin=1}^{400} weighted_{i,j}(bin) \quad (12)$$

where $PhenoMeNal_{i,j}$ is the cumulative memory signal of the EcoPhysF j in year i, and $weighted_{i,j}(bin)$ is the weighted value of the EcoPhysF j in year i, at the given phenological bin.

2.2.5. PhenoMeNals selection and trait modeling

Prior to model training, target traits (e.g., yield, sugar content) are normalized within each site × variety combination using a min–max transformation to the [-1, 1] range. This normalization ensures comparability among traits and sites and stabilizes the regression fitting procedure. Consequently, the regression models are trained and evaluated using the normalized target variables rather than their original measurement units. A stepwise model selection procedure based on the Akaike Information Criterion (AIC) is first applied to each phenological bin across the full set of eight candidate PhenoMeNals (step-AIC function of the MASS R package; Venables and Ripley, 2002). This procedure identifies, for each bin, the most informative combination of PhenoMeNals by computing their selection frequency across all bins and retaining up to a user-defined maximum number for model fitting (default: 4). Multiple linear regression models are then trained and tested with the selected predictors within each phenological bin to forecast the target traits, allowing the temporal mapping of their predictive power along the grapevine cycle. Model fitting is implemented using the caret R package (Kuhn, 2008), employing a leave-one-out cross-validation (LOOCV) strategy to ensure robust assessment. Key accuracy metrics such as the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE) are calculated both for the full model fitting and under the leave-one-out cross--validation (LOOCV) framework to assess predictive performance. To support interpretability and account for parameter uncertainty, regression coefficients are reported together with their 95% confidence intervals. The relative contribution of each predictor is quantified using the Lindeman, Merenda, and Gold (LMG) method (Groemping, 2006), which decomposes the total explained variance (R^2) by averaging over all possible orderings of predictors in the model. In addition, diagnostic outputs are generated for each phenological bin, including the temporal dynamics of correlation between model predictions and phenological development, residual distributions, and accuracy metrics assessing predictive performance.

2.3. The PhenoMeNals R package

The PhenoMeNals workflow has been implemented into an R package (v1.0), consisting of two main functions. First, *phenologyCalibration()* calibrates the phenology model for each site and variety using BBCH phenological observations and weather data (hourly or daily). By default, calibration focuses on the cycle length (°C-days), and BBCH stages 8 (budburst), 61 (start of flowering), 65 (full flowering), 68 (end of flowering), 81 (veraison), and 89 (harvest maturity), as well as chilling thresholds (°C) and requirements (chill days). Users can add other phenophases if observational data are available. The phenology model is parameterized by first estimating the percentage of thermal time accumulated up to BBCH 65 (full flowering), which is used to separate the vegetative period from the reproductive period. Subsequent stages are then expressed as percentages of the remaining thermal time; for instance, if BBCH 81 (veraison) is set at 60%, it is assumed to occur after 60% of the thermal time between BBCH 65 and BBCH 89 has elapsed. Users can set the number of iterations of the simplex optimization. Second, the *runPhenomenals()* function applies the calibrated phenology model to simulate EcoPhysFs and integrate them into the prediction framework. Both functions require weather data (mandatory: latitude, longitude, date/time, temperature, precipitation; optional: relative humidity, solar radiation, wind speed) and either BBCH phenological observations (*phenologyCalibration*) or target trait data (*runPhenomenals*). When daily weather inputs are used, the package reconstructs hourly profiles using astronomical and empirical methods (Supplementary S1). Package users can customize multiple function arguments, including start and end years, site(s), variety(ies), time step of the weather data, the rolling window for EcoPhysF smoothing, the temporal bin size for aggregating the EcoPhysFs, evaluation ranges, the maximum number of PhenoMeNals predictors, and the threshold for multicollinearity filtering (Table 2). In addition, the weighting scheme used to integrate EcoPhysFs into the PhenoMeNal signal is configurable within the package, allowing users to apply alternative weighting strategies (e.g., based solely on the correlation coefficient) in addition to the default formulation used in this study (eq. 11). All inputs are passed directly from R to the C# backend, which manages configuration files automatically and returns outputs to R in structured lists, including (i) processed EcoPhysFs, (ii) intermediate results (autocorrelations, multicollinearity, stepwise selection), and (iii) prediction outputs (normalized and original-scale predictions, model diagnostics, and relative predictor importance). The package is downloadable and fully documented with vignettes on GitHub (https://geomodellab.github.io/phenomenals/).

3. Results

Three distinct applications of the PhenoMeNals framework are presented. First, a proof-of-concept demonstrates the complete workflow, from phenology model calibration to yield prediction, using a single site × variety combination (Fig. 1), with intermediate outputs at each logical step. Second, the framework was extended across 28 sites × year combinations to predict key phenological stages as well as yield and quality traits (sugar content and Brix). This application highlighted correlations between EcoPhysFs and target traits across phenological development, enabling performance benchmarking among varieties, sites, and seasons. Third, we investigated multiple grapevine varieties at the COL site (Table 1), focusing on predicting key yield components (i.e., bunch number and berry weight). This final application emphasized the

Table 2
Summary of input arguments and options for the *phenologyCalibration()* and *runPhenomenals()* functions.

Argument	Description	Default / Example
weather_data	Data frame with hourly or daily weather data (latitude, longitude, date/time, T, P, optionally RH, LW and W)	Required
referenceBBCH	BBCH phenological observations (*phenologyCalibration*)	Required
target_data	Yield, yield components, or quality traits (*runPhenomenals*)	Required
phenomenalsParameters	Nested list of model parameters; includes min, max, default value, and calibration flag	Provided as .rda file
start_year, end_year	First and last year of the simulation or calibration period	2000–2025
sites, varieties	Site(s) and variety(ies) to include	"all", or site/variety name
iterations	Number of iterations for calibration (*phenologyCalibration*)	100
timestep	Time step of weather data ("hourly" or "daily")	"daily"
rolling_window	Rolling window (days) for signal smoothing (*runPhenomenals*)	3
evaluation_range	Percentage range(s) of phenological cycle to compute PhenoMeNals	list(c(100,200), c(300,400))
bin_size	Temporal resolution (percentage of cycle) for aggregating signals (*runPhenomenals*)	1
multicollinearity_threshold	Threshold for excluding highly correlated predictors (*runPhenomenals*)	0.8
max_phenomenals	Maximum number of predictors to retain (*runPhenomenals*)	4
weighting_method	Method used to weight ecophysiological functions by their association with the target trait (eq. 11) ("r_p" = $r(1-p)$; "r" = r).	"r_p"

trajectories of inter-annual memory effects, illustrating their temporal dynamics and relationships with total yield variability. All analyses were performed using the PhenoMeNals R package consistently across case studies: a 3-day rolling window was utilized for smoothing EcoPhysFs, with bin aggregation set to 1 unit, a multicollinearity threshold of 0.9, and selection limited to a maximum of three PhenoMeNals per analysis. Evaluation bin ranges differed according to the specific aims: 0-400 in the initial proof-of-concept (capturing the full annual cycle), 100-200 and 300-400 in the second application (targeting the two growing seasons), and 0-200 and 200-400 in the third application, reflecting the timing of the bunch number and berry weight developmental physiology, respectively.

3.1. Proof-of-concept: grapevine yield for Carmenere in Colli Orientali

The PhenoMeNals pipeline has been applied on CAR variety in COL (Table 1), a variety-site combination with phenological observations and yield data spanning 2006-2019 (Fig. 3). For each year in the dataset (Fig. 3a), the progression through dormancy and the subsequent phenological development have been simulated with RMSE of 7.2 days for bud burst (BBCH 8, occurring around mid-April), 7.1 days for flowering (BBCH 65, early June), 7.3 days for veraison (BBCH 81, early August), and 6.2 days for maturity (BBCH 89, mid-September). The mean seasonal dynamics of the eight EcoPhysFs across the simulation period are shown in Fig. 3b. Temperature suitability (TempF) and light availability (LightF) both exhibited bell-shaped seasonal patterns peaking between late spring and midsummer, with maxima of approximately 0.7 and 0.6, respectively. While both reflect periods of optimal growing conditions, LightF shows a broader distribution across the season, whereas TempF reflects the thermal suitability for grapevine photosynthesis. ColdF displays low values (~0.4) during dormancy and increases after bud break, when cold stress is detrimental to young leaves. The function reaches its maximum (1, no limitation) during the growing season, with a plateau persisting until late autumn, after which ColdF declines again as temperatures fall below the minimum threshold for growth (T_{min} = 10 °C, Fig. 2). Conversely, HeatF remains close to 1 throughout the season, with minor drops to ~0.96 during peak summer, suggesting infrequent but present episodes of air temperature above the heat threshold (T_{heat} = 33.4 °C, Fig. 2). The vapor pressure deficit function (VPDeF) shows a distinct seasonal decline, dropping to around 0.85 during midsummer, indicative of intensified atmospheric evaporative demand. In contrast, the DroughtF value displays higher temporal variability, with short-term fluctuations reflecting the interaction between precipitation events and crop evapotranspiration. WindF is characterized by moderate seasonal variation, with slightly elevated values during summer. DiseaseF presents two seasonal peaks, typically in early and mid-summer, associated with conditions favorable to fungal pathogen development. This EcoPhysF declines later in summer due to the combined effects of high temperatures and reduced leaf wetness, which limit pathogen infections.

The correlation analysis for DroughtF and yield across the two-season reproductive cycle reveals alternating stage-specific sensitivities (Fig. 3c). During ecodormancy of Y–1 (bins ~50–100), correlations are predominantly positive, indicating that higher water availability in winter favors yield in the following year. As the growing season of Y–1 begins (bins 100–200), correlations are weakly positive during differentiation (DIFF), shift negative following inflorescence development (DEV), and then become strongly positive during inflorescence setup (SETUP), indicating that water availability during primordia formation and organogenesis is critical for enhancing yield potential in the subsequent year. A comparable pattern of alternating correlations emerged during the Y0 dormancy and growing season (bins ~200–400). During endodormancy and ecodormancy (200–280), correlations were consistently positive, indicating that ample winter water availability supports yield. A pronounced positive peak at budburst (~300) further underscores the importance of water supply for early vegetative growth. In contrast, correlations shifted negative during flowering (300–320), suggesting that moderate drought stress at anthesis may be favorable for yield. Around veraison (~330–340), correlations reverted to positive, reflecting the sensitivity of ripening onset to water availability. Toward maturity (360–400), correlations weakened and became less consistent, pointing to a diminished influence of water conditions at the end of the cycle.

The EcoPhysFs were then normalized using a sigmoid transformation (eq. 10), and their carry-over effect integrated across years based on established correlations (eq. 11). The Akaike Information Criterion (AIC), applied for each bin, led to identifying the PhenoMeNals from TempF, ColdF and DroughtF as top predictors, with Fig. 3d displaying their interannual trajectories. Marked fluctuations are evident both within and between years. For example, the PhenoMeNal from TempF typically increases during active development in Y0, while the one for ColdF shows stronger cumulative effects in dormancy and early spring. The DroughtF PhenoMeNal exhibits higher interannual variability, with both positive and negative phases depending on the season and year.

Model performance dynamics across the phenological timeline are shown in Fig. 3e, where R^2 and normalized RMSE (nRMSE, computed on yield values normalized from -1 to 1) from leave-one-out cross-validation (LOOCV) quantify predictive accuracy as a function of phenological progression. During dormancy in Y-1 (bins 0-100), R^2 fluctuates considerably, reflecting unstable relationships between predictors and yield, while nRMSE values are high and variable. As the Y-1 growing season begins (~bin 101), R^2 rises above 0.6 and steadily increases through the season, peaking during inflorescence development and shoot growth. nRMSE shows the inverse trend, gradually declining, indicating improved model fit as phenology progresses and weather cues become more informative. In Y0, R^2 values during dormancy (bins 201-300) remain more stable compared to Y-1 but still display moderate variability, albeit around a higher mean (~0.8). The final growth period in Y0 (bins 301-400) is marked by a sharp increase in R^2 to values above 0.9 and a corresponding decline in nRMSE to values below 0.2, indicating optimal performance during late reproductive phases. Fig. 3f displays model-predicted yields for each harvest year, according to the predictive capacity of each phenological bin. The vertical spread of modeled yields across phenological bins reflects the evolving predictive power over the cycle, showing greater accuracy and convergence in later phenological stages, when accumulated memory has substantially informed the predictions.

3.2. Predicting yield and quality on multiple site × variety combinations

Heatmaps of correlations between EcoPhysFs and yield across all site × variety combinations are represented in Fig. 4. Considering DroughtF in Y-1, most varieties in COL, CHA variety in PEN (PEN_CHA) and SAU variety in OYS (OYS_SAU) exhibited positive correlations after DEV, followed by negative correlations after SETUP. In Y0, these were followed by an early positive association after BUD, then a general shift toward negative correlations later in the season; notably, PEN_CHA remains negatively correlated throughout Y0. DiseaseF displays scattered tiles of alternating sign and a slight bias toward negative values in Y0, reflecting highly site- and variety-specific effects. ColdF correlations are sparse and largely episodic in Y-1: most rows for the warmer sites (COL and PEN) are blank (ColdF = 1 throughout the cycle) after spring, whereas cooler sites maintain intermittent correlations around BUD–FLO, indicating ephemeral cold sensitivity. In Y0, negative correlations emerge in the first phases after BUD, indicating a negative association of cold stress with final yield. In LightF, most site × variety combinations showed persistent positive associations across both years after SETUP in Y-1 and after VER in Y0.

TempF is characterized by alternating correlations in Y0, with a prevalence of positive correlations after BUD and before VER, with PEN site showing an opposite trend. Finally, VPDefF reveals a coherent alternating negative-positive pattern in Y0 from BUD to FLO and from

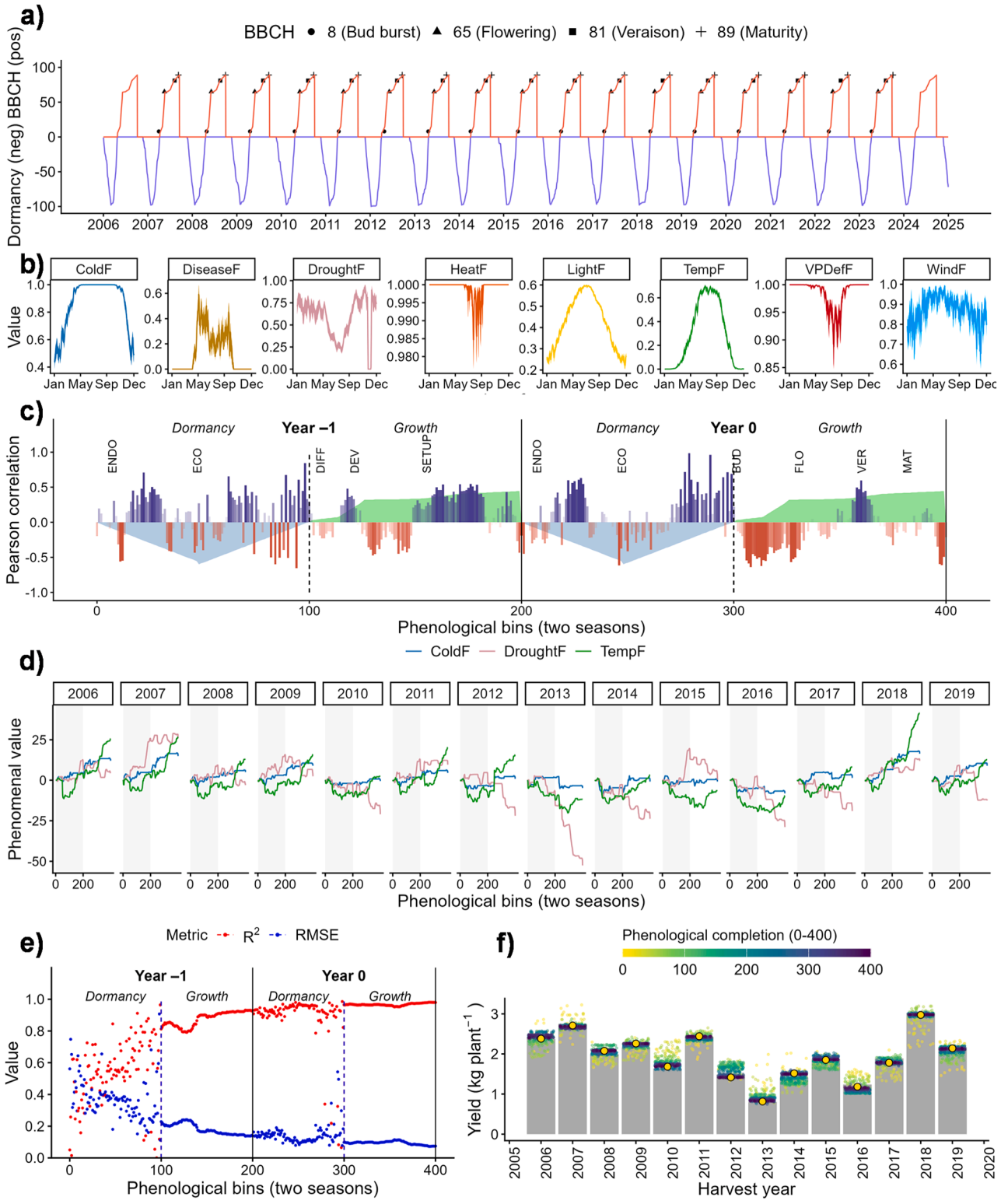


Fig. 3. Proof-of-concept of the PhenoMeNals framework on Carmenere, Colli Orientali. (a) Phenological development expressed as BBCH stage (red, right y-axis) and dormancy index (blue, left y-axis) across the 2006–2025 period; (b) Seasonal dynamics of the eight ecophysiological functions used in the analysis expressed as normalized values (0–1) throughout the annual cycle (day of year on the x-axis); (c) Pearson correlations between yield and DroughtF across two growing seasons (Year-1 and Year 0). Positive and negative correlations are shown in blue and red, respectively. In Year-1, key phenological stages are reported: start of endodormancy (ENDO) and ecodormancy (ECO), primordia differentiation (DIFF) and development (DEV), and inflorescence setup (SETUP); (d) Temporal evolution of the selected PhenoMeNals (ColdF, AridityF, and TempF) across the 2006–2019 period. Grey shading indicates dormancy phases; (e) Model evaluation metrics in leave-one-out cross validation (R^2, red; RMSE, blue) as a function of phenological completion (0–400) across the two years. (f) Observed yield (kg plant^{-1}, grey bars) for each harvest year and corresponding predicted values (colored dots) based on the predictive capacity of each phenological bin. The yellow large dots correspond to the simulated value at bin = 400 from the LOOCV best-performing model.

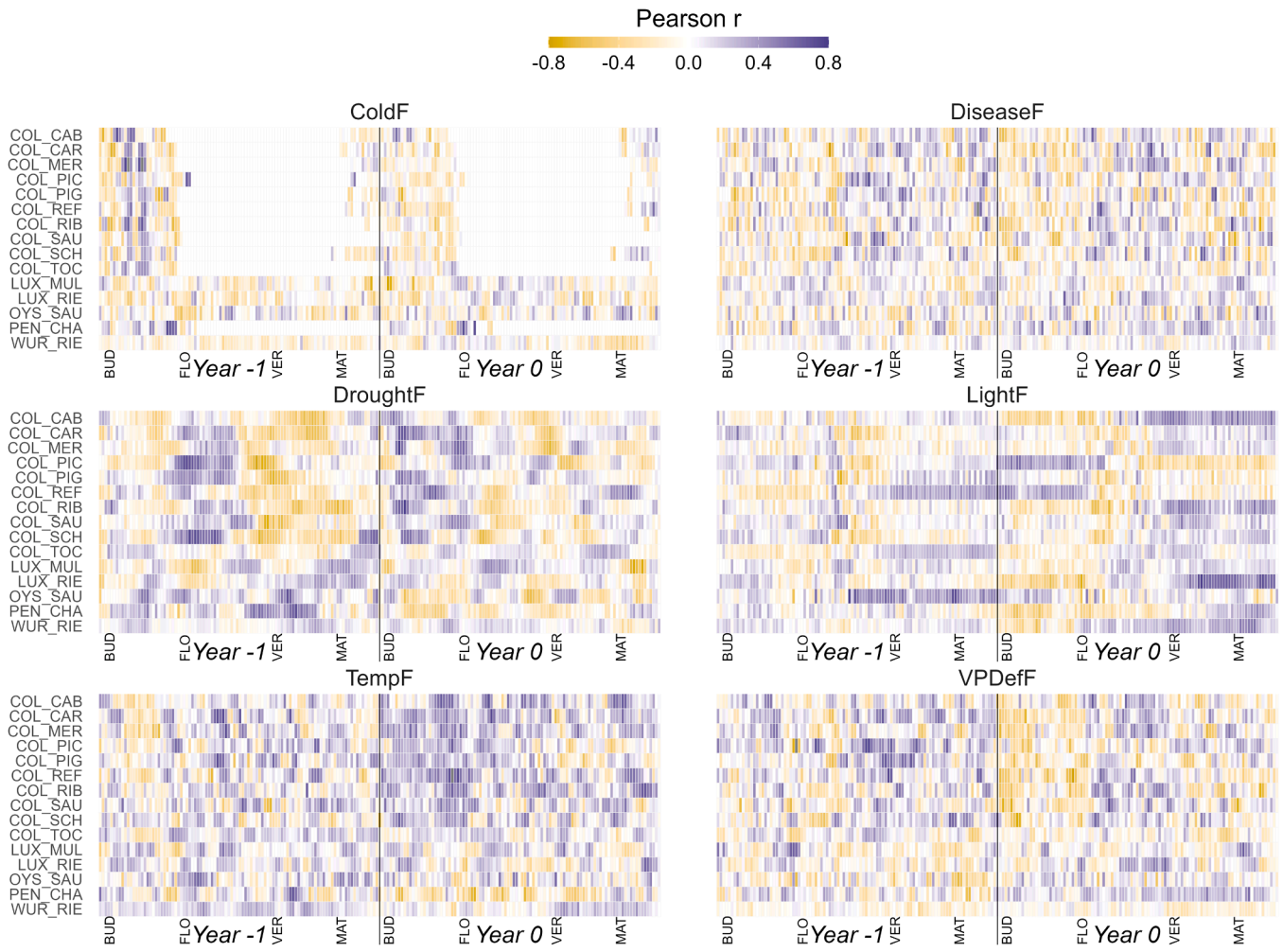


Fig. 4. Correlations between grapevine yield and ecophysiological functions across site × variety combinations (y-axis). Heatmaps show Pearson correlations (blue = positive, yellow = negative) between yield and six ecophysiological functions (ColdF, DiseaseF, DroughtF, LightF, TempF, and VPDefF) calculated on two growing seasons (Year -1 and Year 0). The x-axis represents phenological completion, i.e., previous growing season (Year-1, bins 100-200): primordia differentiation and development (DIFF, DEV), and inflorescence setup (SETUP); Year 0 (bins 300-400): bud break (BUD), flowering (FLO), veraison (VER), maturity (MAT). The vertical line marks the transition between Year-1 and Year 0. Abbreviations for sites and varieties are reported in Table 1.

FLO to VER, respectively.

Building on these correlations, Fig. 5a summarizes the performance of the PhenoMeNals framework in predicting phenological stages, and yield (YIE) and qualitative traits (sugar content, SUG; Brix, BRI). Simulated dates for four key BBCH phases, i.e., BUD (BBCH 8), FLO (BBCH 65), VER (BBCH 81), and MAT (BBCH 89), showed overall agreement with observations across sites, with R^2 ranging from 0.71 to 0.80 and RMSE between 6.8 and 8.7 days. These results indicate comparable accuracy among geographies, including OYS, whose growth cycle was standardized by adjusting the day of the year (6-month shifting) to allow comparison between the two hemispheres. Flowering and maturity stages obtained best predictive performances, denoting the model ability to accurately capture both the onset and culmination of the reproductive cycle. Fig. 5b presents the evolution of predictive performance (R^2 and RMSE) for both full-model fitting and leave-one-out cross-validation (LOOCV) across the phenological timeline, whereas Table 3 reports accuracy metrics at bin 400 (maturity in Y0). In most site × variety combinations, the correlation strength increased progressively from the beginning of Y0, with R^2 stabilizing toward the end of the season, indicating that predictive power accumulated as more of the phenological cycle was captured. The gap between full-model and LOOCV performance remained narrow across bins, showing that the models generalized well beyond the calibration data. At phenological bin 400, yield (YIE) predictions achieved consistently high accuracy across sites and varieties (Table 3). Here, nRMSE values are reported on a normalized scale ranging from –1 to 1, where values closer to 0 indicate better predictive performance. Full models typically reached R^2 values above 0.90 with RMSE below 0.12, while LOOCV results were only slightly lower, with R^2 values above 0.85 and RMSE below 0.20 in most cases. Best results were obtained for OYS_SAU ($R^2 = 0.98$ full, 0.95 LOOCV; RMSE = 0.06 and 0.09), while COL varieties also showed consistently strong performance. By contrast, PEN_CHA and WUR_RIE displayed lower accuracy, with LOOCV R^2 around 0.82 and 0.76, respectively. Only a few site × variety combinations showed larger discrepancies, suggesting site-specific variability or greater sensitivity to interannual fluctuations.

For sugar-related traits (SUG, BRI), the performance was more variable. In COL varieties, both YIE and SUG achieved comparable accuracy (mean LOOCV $R^2 \approx 0.88$; normalized RMSE, nRMSE ≈ 0.18–0.19). WUR_RIE had the weakest sugar predictions (LOOCV $R^2 = 0.72$; nRMSE = 0.24). NAP varieties performed well for Brix (mean LOOCV $R^2 = 0.84$; nRMSE = 0.22), while PEN showed a decline in LOOCV accuracy relative to the full model, indicating possible overfitting or greater interannual variability. Across the 28 site × variety combinations, 11 achieved LOOCV R^2 above 0.85 and 20 exceeded 0.75, demonstrating that most models reached stable and reliable predictive accuracy by bin

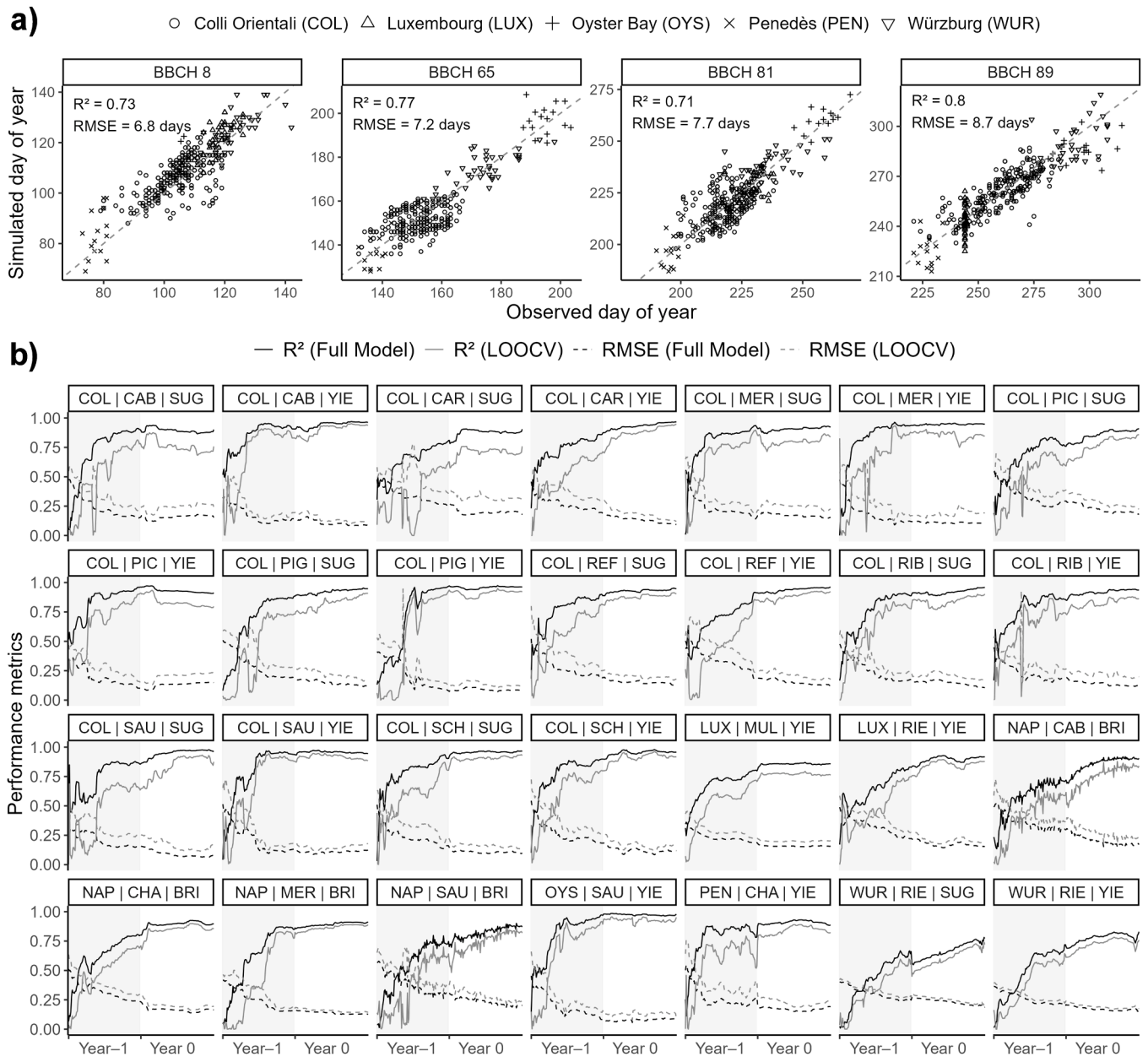


Fig. 5. Model evaluation for phenology, yield, and quality predictions across sites and varieties.
(a) Simulated versus observed day of year for the four reference BBCH phenological stages: bud burst (BBCH 8), flowering (BBCH 65), veraison (BBCH 81), and maturity (BBCH 89). Each point represents a site × variety × year combination, with symbols corresponding to the five study sites. The dashed line indicates perfect agreement. (b) Performance metrics for yield (YIE), sugar content (SUG), and Brix (BRI) predictions across site × variety combinations. Solid lines represent R^2 and dashed lines RMSE, with the full model shown in black and leave-one-out cross-validation (LOOCV) in grey. The x-axis represents the standardized phenological completion scale (Year-1, grey, and Year 0, white). Panel labels denote site, variety, and predicted trait (e.g., COL | CAB | SUG = Colli Orientali | Cabernet | Sugar content).

400. Overall, yield predictions were consistently more robust than sugar-related traits, although both targets showed improved performance as the season progressed.

3.3. Focus on yield components: bunch number and berry weight

In the third case study focusing on bunch number and berry weight variability for varieties in COL, the PhenoMeNals temporal profiles highlighted strong seasonal and inter-annual patterns (Fig. 6 for bunch number, Supplementary S4 for berry weight). Observed bunch number varied strongly among years (Fig. 6a), with particularly low values in 2009, 2013 and 2017 and high values in 2006, 2007, 2018 and 2019. Examination of the Y-1 season (Fig. 6b) revealed clear temporal associations of the PhenoMeNals with the target trait. Years characterized by above-average bunch number were consistently preceded by favorable conditions, specifically prolonged positive correlations in the PhenoMeNals from TempF (approximately +10 to +15) and LightF (approximately +15 to +20), combined with limited negative excursions in stress-related functions such as ColdF, DroughtF, and VPDefF. Conversely, years exhibiting notably reduced bunch numbers (2009, 2013, and 2017) coincided with distinctly adverse Y-1 PhenoMeNals trajectories. For instance, 2013, representing the lowest observed bunch number, was preceded by substantial negative deviations in multiple functions: severe negative spikes in PhenoMeNals from HeatF, DroughtF

Table 3
Predictive performance of models at phenological bin 400 (maturity in the harvest year) for all site × variety combinations. Reported are normalized RMSE and R^2 values for the full model fit and leave-one-out cross-validation (LOOCV). YIE = yield; SUG = sugar-related traits (sugar content or Brix). Acronyms for sites and varieties are explained in Table 1.

Target trait	Site	Variety	nRMSE		R^2	
			Full model	LOOCV	Full model	LOOCV
YIE	COL	CAB	0.09	0.115	0.964	0.94
		CAR	0.095	0.121	0.967	0.947
		MER	0.108	0.193	0.948	0.846
		PIC	0.149	0.227	0.913	0.798
		PIG	0.087	0.124	0.961	0.921
		REF	0.123	0.176	0.961	0.923
		RIB	0.123	0.199	0.937	0.863
		SAU	0.123	0.179	0.945	0.885
		SCH	0.118	0.171	0.957	0.916
	LUX	MUL	0.158	0.204	0.857	0.761
		MUL	0.154	0.187	0.918	0.879
	OYS	SAU	0.058	0.087	0.979	0.952
	PEN	CHA	0.192	0.242	0.885	0.818
	WUR	RIE	0.146	0.174	0.83	0.76
SUG	COL	CAB	0.153	0.243	0.902	0.758
		CAR	0.177	0.282	0.905	0.759
		MER	0.155	0.226	0.923	0.839
		PIC	0.179	0.225	0.911	0.86
		PIG	0.122	0.16	0.948	0.909
		REF	0.131	0.178	0.948	0.905
		RIB	0.104	0.165	0.96	0.902
		SAU	0.077	0.156	0.967	0.875
		SCH	0.099	0.139	0.964	0.929
	WUR	RIE	0.189	0.21	0.773	0.722
	NAP	CAB	0.171	0.222	0.905	0.84
		CHA	0.164	0.196	0.897	0.853
		MER	0.125	0.142	0.914	0.89
		SAU	0.204	0.241	0.87	0.819

and ColdF (~ -20), accompanied by significant negative troughs in PhenoMeNals from VPDefF and LightF (i.e., the only year with negative value), and suppressed TempF dynamic.

Similarly, the low bunch number in 2009 was preceded by pronounced negative values for the PhenoMeNals from ColdF (~ -10), DroughtF (~ -5), and limited positive values for LightF. The decline observed in 2017 was likewise anticipated by negative values of the PhenoMeNals from DroughtF (~ -15) and LightF. The DiseaseF PhenoMeNal exhibited v-shaped patterns, whose final values seem correlated with lower bunch number, as exemplified by the conditions leading up to 2013 and 2014. Although the PhenoMeNal from HeatF also displayed noticeable fluctuations in some years, its temporal patterns were less consistently associated with bunch number variability compared with TempF and LightF. Overall, sustained positive values in these two PhenoMeNals, along with the absence of significant negative stress indicators (ColdF, DroughtF, VPDefF), consistently preceded high bunch numbers, whereas pronounced negative deviations in these scaled EcoPhysFs consistently reduced this trait.

Observed and simulated bunch number and berry weight across ten grapevine varieties at COL site is shown in Fig. 7. For bunch number, predictions closely matched the observed inter-annual variability, resulting in high R^2 values consistently above 0.8 in cross-validation (R^2 LOCCV) and frequently near or above 0.9 for full models. Berry weight also showed strong predictive performance, although slightly lower than for bunch number, with R^2 typically ranging from 0.82 to 0.94 and cross-validation R^2 values consistently above 0.7. Notably, varieties differed in their pattern of variability and prediction accuracy. For instance, varieties such as PIC, PIG, and SCH displayed exceptional accuracy in simulations for both yield components. Conversely, TOC and CAR exhibited lower predictive accuracy, particularly for berry weight (R^2 LOCCV around 0.7-0.73), indicating that these varieties may be influenced by additional factors not captured by the selected EcoPhysFs.

4. Discussion

4.1. A new approach to yield and quality forecasting

The foundation of the PhenoMeNals framework lies in reproducing plant phenology, which is the cornerstone of any yield prediction logic, for grapevine (Petrie and Clingeleffer, 2005) as well as for any other crops (Pei et al., 2025). Phenology delineates the occurrence of all physiological processes during the grapevine cycle, from floral initiation and early inflorescence development in the previous season, through dormancy and bud differentiation, up to anthesis and ripening in the harvest year (Carmona et al., 2008). Because reproductive potential is set across this two-year timeline, weather conditions from both seasons inherently shape yield and quality traits, making it essential to account for carry-over effects (Faralli et al., 2024). The phenology simulator used in this study is interfaced with the widely used simplex optimization algorithm (e.g., Valeriano et al., 2021; Kunecký et al., 2024), which minimizes the errors (days) in simulating phenological observations coded via BBCH scale. The same dormancy model has been adopted for various perennial tree crops including olive (Didevarasl et al., 2025), kiwi (Kim and Koh, 2019), and apple (Ahn et al., 2024). It conceptually delineates the progression through endodormancy and ecodormancy using a simple but robust sequential approach based on chill/anti-chill days (Cesaraccio et al., 2004). The forcing component relies on a well-established thermal time accumulation method (Yan and Hunt, 1999) to simulate consecutive phases aligned with the BBCH scale, consistently with other phenological studies (Molitor et al., 2020). The PhenoMeNals framework obtained solid performances across six sites located in heterogenous climatic regions covering extended time series and involving 17 grapevine varieties. Achieving an RMSE between 6 and 9 days for critical phenophases (i.e., bud break, flowering, veraison and maturity), our results align with recent literature on grapevine phenology modelling: Reis et al. (2020) reported an average RMSE of approximately 7 days, while Zapata et al. (2017) obtained values between 5 and 12 days (excluding maturity) across multi-year studies involving extensive varietal datasets in Portugal and the United States. Similarly, Omazić et al. (2023) reported RMSE values ranging from 5 to 17 days for four grapevine varieties over a long-term historical dataset (1961-2020) in Croatia.

Existing approaches for grapevine yield forecasting range from purely empirical models, which use seasonal or phenology-centered climatic predictors via regression (e.g., Santos et al., 2020) or machine learning (e.g., Andrade et al., 2023), to complex process-based simulators such as STICS (Fraga et al., 2015), APSIM (Zhu et al., 2021) and GrapeML (Leolini et al. 2018), which however demand extensive parameterization. Following the principles introduced by Massano et al. (2023), who used ecoclimatic indices aligned with phenophases to explain grapevine yield variability, we developed a hybrid approach that merges the interpretability of process-based modeling with the flexibility of data-driven approaches. However, unlike Massano et al. (2023), our EcoPhysFs are computed throughout the phenological timeline. Parameters and functional shapes were derived from literature to ensure physiological relevance and robustness (Supplementary S1). Since considerable uncertainty emerged across studies, we set the mean values as default parameters, while computing standard deviations to allow users adjusting them within plausible ranges. For example, frost damage thresholds (T_{cold}) have been reported to vary from -4.5 °C to -0.5 °C, whereas cardinal temperatures were surprisingly different across studies (T_{min} 5-12 °C, T_{opt} 20-35 °C, T_{max} 30-35 °C). Each EcoPhysF is then normalized within each phenological bin using long-term percentile distributions to standardize its relative expression across years. This normalization step makes the approach phenology-aware, differently from calendar-based window analyses used by other weather-cue identification methods such as climate sensitivity profiles (Simmonds et al., 2019), P-spline regression (Roberts et al., 2015) and sliding-window (van de Pol et al., 2016). Each scaled EcoPhysF is subsequently

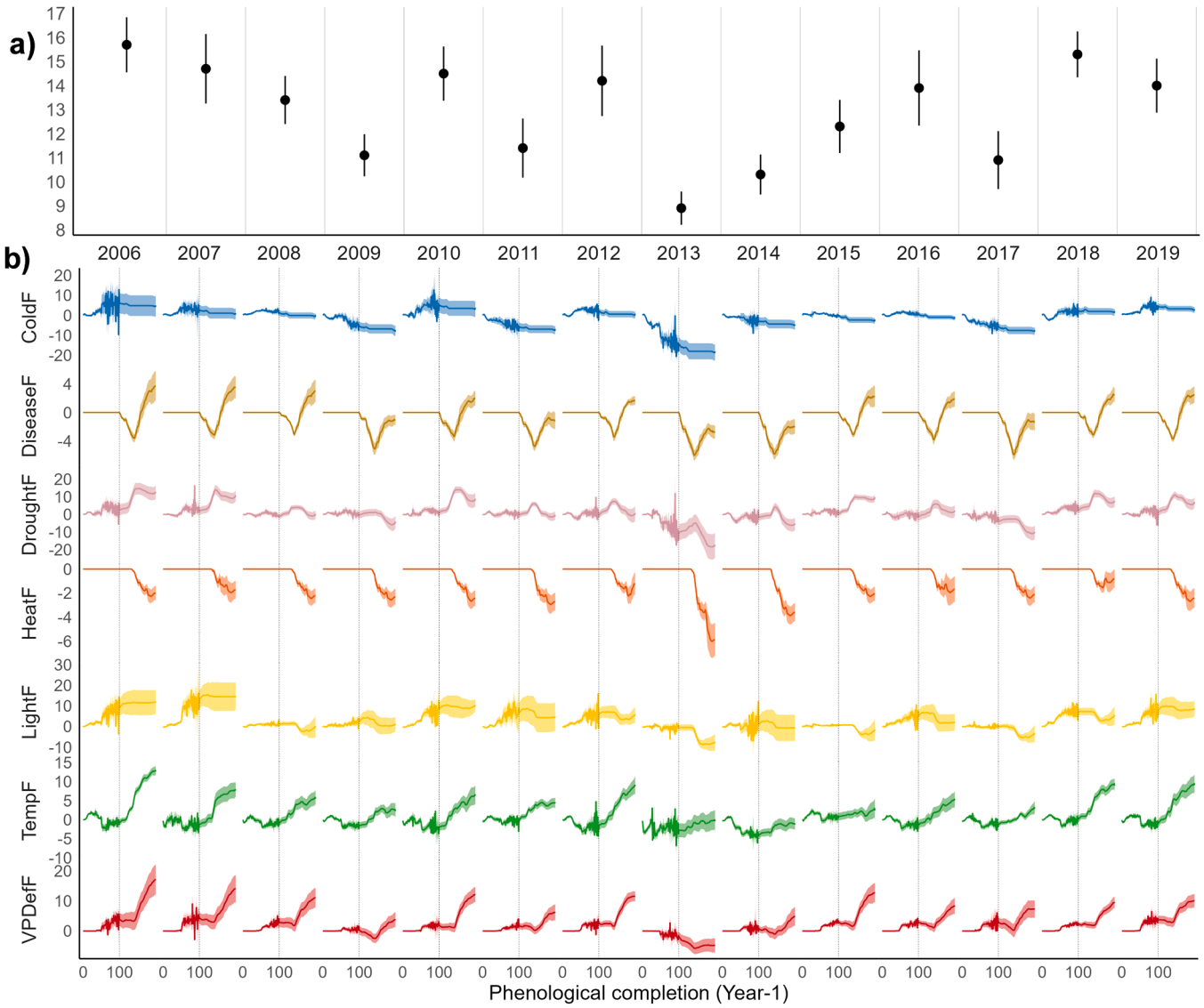


Fig. 6. Actual bunch number and PhenoMeNals temporal dynamics for their prediction in Colli Orientali; (a) Observed bunch number (mean ± SE) from 2006 to 2019. (b) Temporal evolution of the memory signals of the functions: ColdF (blue), DiseaseF (yellow), DroughtF (pink), HeatF (orange), LightF (yellow), TempF (green), and VPDefF (red). Each function is expressed on the standardized phenological scale referred to the previous growing season (Year-1, x-axis; 0-200). Positive and negative values represent the relative direction and intensity of the association with bunch number. The vertical grey line separates the dormancy from the growing season.

weighted based on the strength and significance of its correlation with the target trait. The seasonal under- or over-expression of the scaled EcoPhysF is cumulatively integrated across the entire phenological timeline to form what we defined 'memory signals', i.e.. the PhenoMeNals. Therefore, each trajectory reflects how weather conditions progressively contribute to shaping yield and quality traits, explicitly accounting for carry-over effects in contrast to relationships centered on specific phenophases (e.g., Zhu et al., 2020; Fraga et al., 2024). The underlying idea is that the predictive capacity improves as it becomes progressively informed by the influence of past conditions, forming a memory-driven continuum which unfolds as the season progresses.

4.2. *Predictive ability and relation with reproductive physiology*

The performance of the PhenoMeNals framework was tested across various sites, varieties, and multiple target yield and quality traits. The availability of reference dataset to validate the framework was surprisingly limited, as they were rarely released with the original publications; therefore, except for Bai et al. (2022), most data were digitized directly from seminal papers. For the COL site, we benefited from a unique series of annual publications that provided complete datasets, which were likewise digitized (https://www.colliorientali.com/). Weather data were obtained from a nearby regional station for COL and from public global or national databases for the other sites (Table 1). While the use of non-site-specific weather data strengthened the generalizability of the framework, it may also have introduced additional uncertainty and reduced predictive performance. The resulting dataset, comprehensive of weather, phenology and yield-related data is released along with this paper (Supplementary S2). When compared with existing yield forecasting studies, the accuracy of the PhenoMeNals framework proved competitive and often superior. Previous process-based and regression approaches typically reached correlations or R^2 values in the range of 0.3-0.7 (e.g., Leolini et al., 2018; Bock et al., 2013; Parker et al., 2020), whereas our framework consistently exceeded 0.8 and approached 0.9 in cross-validation. Only the stage-specific models by Molitor and Keller (2016) reported comparable or higher performance, though their analysis excluded years with extreme frost events, while our framework accommodates such anomalies. More broadly, PhenoMeNals performed

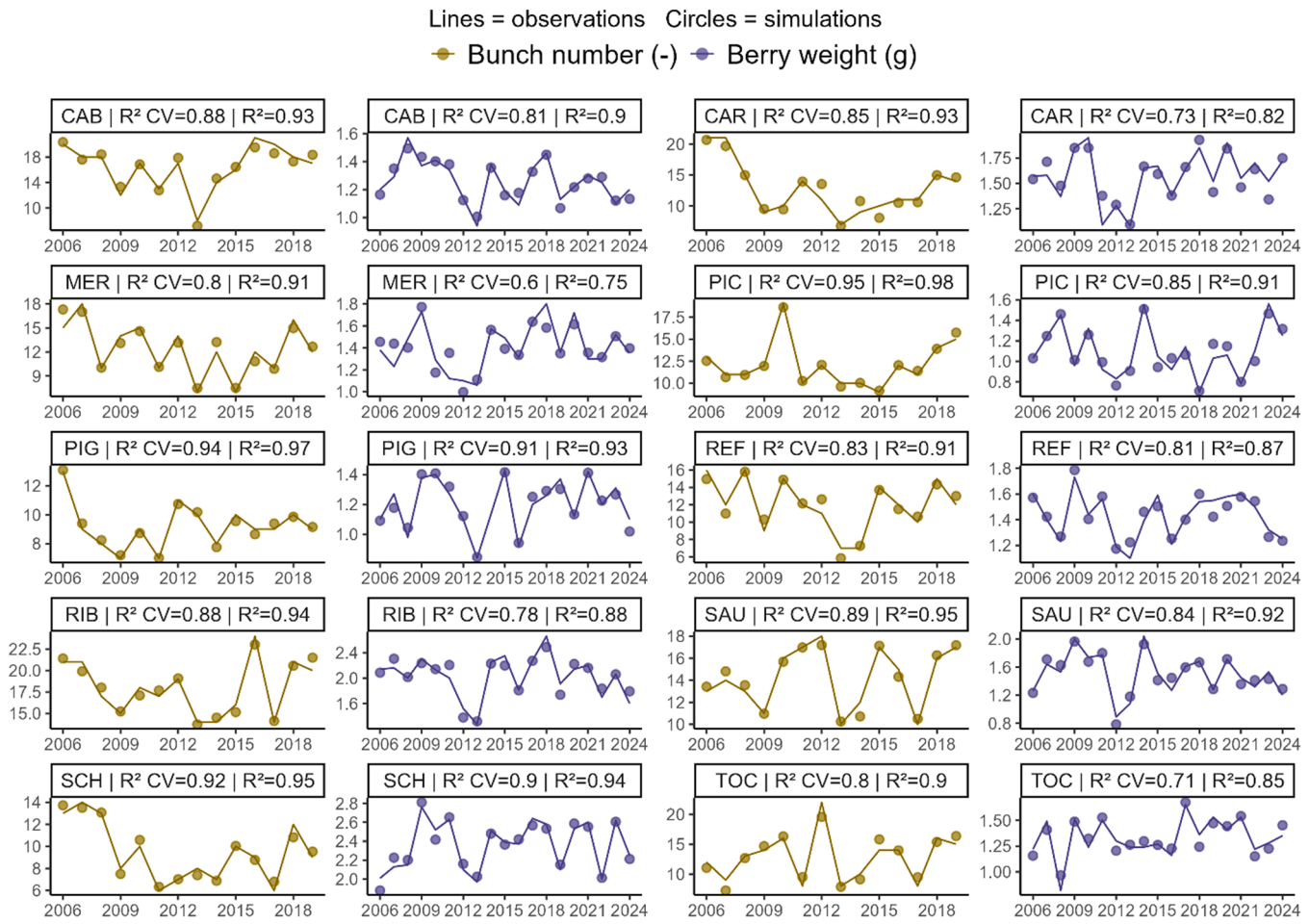


Fig. 7. Observed versus simulated bunch number and berry weight at Colli Orientali (COL). Observed (lines) and simulated (circles) values for bunch number (green; left panels) and berry weight (blue; right panels) from 2006 to 2018 (bunch number) and 2006 to 2021 (berry weight) across ten varieties. Abbreviations for varieties are explained in Table 1. Model performance is indicated by coefficients of determination (R^2) calculated both from calibration (R^2) and leave-one-out cross-validation (R^2 CV).

in line with or better than other established methods, including multi-stage models (de la Fuente et al., 2015), machine-learning algorithms such as Random Forest or Cubist (Andrade et al., 2023), and even advanced computer vision–based forecasting (Palacios et al., 2022; Massano et al., 2023).

4.3. Strengths and weaknesses

Beyond predictive accuracy, the PhenoMeNals provide unique advantages in its ability to progressively improve performance throughout the growing season and to offer physiologically meaningful insights. This approach enables inspecting weather cue sensitivities according to phenophases, thereby linking prediction with physiological understanding. For example, the framework corroborates established knowledge (Laurent et al., 2021) that the influence of external factors on yield development depends on (i) the factor considered (e.g., temperature, abiotic and biotic stresses), (ii) the timing of the influence relative to grapevine developmental stages (e.g., temperature around budbreak or bloom), (iii) the intensity of the influence (e.g., mild vs. extreme temperatures), and (iv) the duration of the exposure (e.g., short vs. prolonged events). Moreover, by capturing trajectory effects, our framework moves beyond the punctual indicators used in most previous studies (e.g., Guilpart et al., 2014; Petrie and Clingeleffer, 2005). Detailed analysis of PhenoMeNals trajectories highlighted critical phenological windows associated with yield-determining events. These windows match established physiological milestones, such as bud differentiation and inflorescence primordia formation occurring predominantly during the prior growing season (Vasconcelos et al., 2009). Thus, the framework not only delivers robust predictions but also provides biologically meaningful insights into the temporal dynamics of yield formation. An additional advantage of the PhenoMeNals framework with respect to current tools lies in its user-oriented flexibility. Parameters calibrated within this study are provided alongside the software package, allowing users to apply it without necessarily requiring new BBCH observations for the same varieties. Further, the package offers users alternative strategies for parameterization to address varying data availability scenarios: applying default values averaged across all existing datasets or using new BBCH observations for phenology recalibration.

However, several limitations should be acknowledged. In this study, the framework was primarily driven by publicly available weather data to demonstrate its applicability across heterogeneous sites and data availability scenarios. However, such datasets may not capture the fine-scale microclimatic variability experienced by grapevine canopies, which can differ substantially from standard meteorological measurements due to local topography, site exposure, and management practices such as trellising and canopy architecture. These factors can alter temperature, radiation, and humidity conditions at the plant level and thus influence phenological development and yield formation (Fonseca et al., 2025). However, the PhenoMeNals framework is designed to flexibly incorporate weather data from different sources and temporal resolutions (e.g., daily or hourly), allowing future applications to integrate

vineyard-level meteorological measurements and potentially improve the representation of local microclimatic conditions. The selected sites and datasets, while extensive, do not fully represent global viticultural diversity, underscoring the need for broader validation. In many cases, yield, quality, and phenological data had to be digitized from published figures, which may introduce inaccuracies compared to original raw datasets. In addition, parameter uncertainty also remains an issue, as the parameters of the EcoPhysFs derived from literature (Supplementary S3) exhibit considerable variability and may not fully capture site- or variety-specific conditions. Furthermore, the correlation-weighted approach assumes linear relationships between memory signals and traits, which may oversimplify complex interactions or threshold effects (Webber et al. 2022). Finally, the availability and completeness of records varied among sites and varieties, which could introduce bias and limit the robustness of comparisons across environments.

5. Conclusions

Despite numerous approaches for grapevine yield and quality prediction have been proposed to date, three main areas remain uncovered: 1) operational relevance, 2) accounting for temporal dynamic of yield development, and 3) adopting a site-specific forecasting approach (see Laurent et al., 2021). The PhenoMeNals framework tries to address these gaps by releasing a predicting workflow which employs phenology simulation as the foundation for EcoPhysFs, which are processed by statistical methods to derive dynamic weather 'mnemonic' effects. The methodology is released as an R package, providing researchers and practitioners with an open-source tool fostering reuse and collaborative progress as well as meaningful comparisons across sites. The framework has been tested across a diverse range of environments and varieties, demonstrating promising predictive performance, yet further validation in other viticultural contexts is essential to assess its generalizability. Beyond predictive strengths, the PhenoMeNals provides physiologically meaningful insights by allowing weather cue sensitivity to be inspected on specific phenophases. We propose it as an interpretable tool for operational yield and quality forecasting, with the expectation that continued development will strengthen its potential for decision-support to the wine industry.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to improve the flow and clarity of the English language. All content generated with its assistance was carefully reviewed and edited by the authors, who take full responsibility for the final published article.

CRediT authorship contribution statement

Simone Bregaglio: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Sofia Bajocco:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Simone Bregaglio, Sofia Bajocco reports administrative support, article publishing charges, and travel were provided by European Commission. Simone Bregaglio, Sofia Bajocco reports administrative support, article publishing charges, and travel were provided by Italian Ministry of Agriculture, Food Sovereignty and Forestry. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.agrformet.2026.111202.

Data availability

All data used in this paper are provided as electronic tables in Supplementary S2 (Excel format)

References

Ahn, M.I., Yang, H.J., Yun, S.C., 2024. Development of K-Maryblyt for fire blight control in apple and Pear Trees in Korea. Plant Pathol. J. 40, 290–298. https://doi.org/10.5423/PPJ.OA.02.2024.0038.

Alikadic, A., Pertot, I., Eccel, E., Dolci, C., Zarbo, C., Caffarra, A., De Filippi, R., Furlanello, C., 2019. The impact of climate change on grapevine phenology and the influence of altitude: a regional study. Agric. For. Meteorol. 271, 73–82. https://doi.org/10.1016/j.agrformet.2019.02.030.

Alonso, R., Muñoz, F., Bottini, R., Piccoli, P., Berli, F.J., 2024. Effects of wind exposure and deficit irrigation on vegetative growth, yield components and berry composition of malbec and cabernet sauvignon. Plants 13, 1292. https://doi.org/10.3390/plants13101292.

Andrade, C.B., Moura-Bueno, J.M., Comin, J.J., Brunetto, G., 2023. Grape yield prediction models: approaching different machine learning algorithms. Horticulturae 9, 1294. https://doi.org/10.3390/horticulturae9121294.

Bai, H., Gambetta, G.A., Wang, Y., Kong, J., Long, Q., Fan, P., Duan, W., Liang, Z., Dai, Z., 2022. Historical long-term cultivar×climate suitability data to inform viticultural adaptation to climate change. Sci. Data 9, 271. https://doi.org/10.1038/s41597-022-01367-6.

Barriguinha, A., de Castro Neto, M., Gil, A., 2021. Vineyard yield estimation, prediction, and forecasting: a systematic literature review. Agronomy 11, 1789. https://doi.org/10.3390/agronomy11091789.

Bock, A., Sparks, T.H., Estrella, N., Menzel, A., 2013. Climate-induced changes in grapevine yield and must sugar content in franconia (Germany) between 1805 and 2010. PLoS. One. https://doi.org/10.1371/journal.pone.0069015.

Bregaglio, S., Rossi, E., Ascari, L., Mongiano, G., Del Cavallo, E., Bajocco, S., Manici, L. M., Pepe, A.G., Bassi, C., Tiso, R., Pietrangeli, F., Cattaneo, G., Nigro, C., Gerardi, M. S., Bussotti, S., Sanchioni, A., Tognetti, D., Sandra, M., De Lillo, I., Framarin, P., Di Ferdinando, S., Bugiani, R., 2025. Releasing the octoPus, an open-source digital tool to promote Integrated Pest Management. Smart Agric. Technol. 11, 101015. https://doi.org/10.1016/j.atech.2025.101015.

Bregaglio, S., Carriero, G., Calone, R., Romano, M., Bajocco, S., 2024. Playing a crop simulation model using symbols and sounds: the 'mandala'. In. Silico Plants. 6, diad023. https://doi.org/10.1093/insilicoplants/diad023.

Bregaglio, S., Savian, F., Raparelli, E., Morelli, D., Epifani, R., Pietrangeli, F., Nigro, C., Bugiani, R., Pini, S., Culatti, P., Tognetti, D., Spanna, F., Gerardi, M., Delillo, I., Bajocco, S., Fanchini, D., Fila, G., Ginaldi, F., Manici, L.M., 2022. A public decision support system for the assessment of plant disease infection risk shared by Italian regions. J. Environ. Manag., 101015 https://doi.org/10.1016/j.jenvman.2022.115365.

Cameron, W., Petrie, P.R., Barlow, E.W.R., 2022. The effect of temperature on grapevine phenological intervals: Sensitivity of budburst to flowering. Agric. For. Meteorol. 315, 108841. https://10.1016/j.agrformet.2022.108841.

Carmona, M.J., Chaïb, J., Martínez-Zapater, J.M., Thomas, M.R., 2008. A molecular genetic perspective of reproductive development in grapevine. J. Exp. Bot. 59, 2579–2596. https://doi.org/10.1093/jxb/ern160.

Cesaraccio, C., Spano, D., Snyder, R.L., Duce, P., 2004. Chilling and forcing model to predict bud-burst of crop and forest species. Agric. For. Meteorol. 126, 1–13. https://doi.org/10.1016/j.agrformet.2004.03.002.

Confalonieri, R., Bregaglio, S., Acutis, M., 2012. Quantifying plasticity in simulation models. Ecol. Model. 225, 159–166. https://doi.org/10.1016/j.ecolmodel.2011.11.022.

de la Fuente, M., Linares, R., Baeza, P., Miranda, C., Lissarrague, J.R., 2015. Comparison of different methods of grapevine yield prediction in the time window between

fruitset and veraison. OENO One 49, 27–35. https://doi.org/10.20870/oeno-one.2015.49.1.96.
Delelee, L.M., Woodill, A.J., Skinkis, P.A., 2025. Climate change projections indicate shifts in phenology for willamette valley pinot noir. Am. J. Enol. Vitic. 76, 0760003. https://doi.org/10.5344/ajev.2024.24033.
Didevarasl, A., Costa-Saura, J.M., Spano, D., Deiana, P., Snyder, R.L., Rechid, D., Bülow, K., Mulas, M., Nieddu, G., Trabucco, A., 2025. The phenological phases of early and mid-late budbreak olive cultivars in a changing future climate over the Euro-Mediterranean region. Eur. J. Agron. 168, 127658. https://doi.org/10.1016/j.eja.2025.127658.
Duchêne, É., 2016. How can grapevine genetics contribute to the adaptation to climate change? Oeno One 50, 113–124. https://doi.org/10.20870/oeno-one.2016.50.3.98.
Faralli, M., Mallucci, S., Bignardi, A., Varner, M., Bertamini, M., 2024. Four decades in the vineyard: the impact of climate change on grapevine phenology and wine quality in northern Italy. OENO One 58. https://doi.org/10.20870/oeno-one.2024.58.3.8083.
Fonseca, A., Cruz, J., Fraga, H., Andrade, C., Valente, J., Alves, F., Neto, C., Flores, R., Santos, J.A., 2025. Enhancing climate change impact assessment in viticulture by resolving microclimates. Eur. J. Agron. 170, 127740. https://doi.org/10.1016/j.eja.2025.127740.
Fraga, H., Costa, R., Moutinho-Pereira, J., Correia, C.M., Dinis, L.T., Gonçalves, I., Silvestre, J., Eiras-Dias, J., Malheiro, A.C., Santos, J.A., 2015. Modeling phenology, water status, and yield components of three Portuguese grapevines using the STICS crop model. Am. J. Enol. Vitic. 66, 482–491. https://doi.org/10.5344/ajev.2015.15031.
Fraga, H., Freitas, T.R., Moriondo, M., Molitor, M., Santos, J.A., 2024. Determining the climatic drivers for wine production in the Côa Region (Portugal) using a machine learning approach. Land 13, 749. https://doi.org/10.3390/land13060749.
Freeman, B.M., Kliewer, W.M., Stern, P., 1982. Influence of windbreaks and climatic region on diurnal fluctuation of leaf water potential, stomatal conductance, and leaf temperature of grapevines. Am. J. Enol. Vitic. 33, 233–236. https://doi.org/10.5344/ajev.2009.60.4.411.
Ghiglieno, I., Carlin, S., Cola, G., Vrhovsek, U., Valenti, L., Garcia-Aloy, M., Mattivi, F., 2023. Impact of meteorological conditions, canopy shading and leaf removal on yield, must quality, and norisoprenoid compounds content in Franciacorta sparkling wine. Front. Plant Sci. 14. https://doi.org/10.3389/fpls.2023.1125560.
González-Fernández, E., Piña-Rey, A., Fernández-González, M., Aira, M.J., Rodríguez-Rajo, F.J., 2020. Prediction of grapevine yield based on reproductive variables and the influence of meteorological conditions. Agronomy 10, 714. https://doi.org/10.3390/agronomy10050714, 2020.
Groemping, U., 2006. Relative importance for linear regression in R: the package relaimpo. J. Stat. Softw. 17, 1–27. https://doi.org/10.18637/jss.v017.i01.
Guilpart, N., Metay, A., Gary, C., 2014. Grapevine bud fertility and number of berries per bunch are determined by water and nitrogen stress around flowering in the previous year. Eur. J. Agron. 54, 9–20. https://doi.org/10.1016/j.eja.2013.11.002.
He, L., Fang, W., Zhao, G., Wu, Z., Fu, L., Li, R., Majeed, Y., Dhupia, J., 2022. Fruit yield prediction and estimation in orchards: A state-of-the-art comprehensive review for both direct and indirect methods. Comput. Electron. Agric. 195, 106812. https://doi.org/10.1016/j.compag.2022.106812.
Huang, M., Aine, C.J., Supek, S., Best, E., Ranken, D., Flynn, E.R., 1998. Multi-start downhill simplex method for spatio-temporal source localization in magnetoencephalography. Electroencephalogr. Clin. Neurophysiol./Evoked Potentials Sect. 108, 32–44. https://doi.org/10.1016/S0168-5597(97)00091-9.
Jarvis, P.G., Monteith, J.L., Weatherley, P.E., 1976. The interpretation of the variations in leaf water potential and stomatal conductance found in canopies in the field. Philos. Trans. R. Soc. Lond. B. Biol. Sci. 273, 593–610. https://doi.org/10.1098/rstb.1976.0035.
Kim, K.H., Koh, Y.J., 2019. An integrated modeling approach for predicting potential epidemics of bacterial blossom blight in kiwifruit under climate change. Plant Pathol. J. 35, 459–472. https://doi.org/10.5423/PPJ.OA.05.2019.0140.
Kuhn, M., 2008. Building predictive models in R using the caret package. J. Stat. Softw. 28, 1–26. https://doi.org/10.18637/jss.v028.i05.
Kunecký, J., Štembera, V., Kloiber, M., Hataj, M., Sebera, V., 2024. Optimization of CZM parameters of Norway spruce tested in mode I using multistart simplex method. Eng. Fract. Mech. 310, 110430. https://doi.org/10.1016/j.engfracmech.2024.110430.
Laurent, C., Oger, B., Taylor, J.A., Scholasch, T., Metay, A., Tisseyre, B., 2021. A review of the issues, methods and perspectives for yield estimation, prediction and forecasting in viticulture. Eur. J. Agron. 130, 126339. https://doi.org/10.1016/j.eja.2021.126339.
Leolini, L., Bregaglio, S., Moriondo, M., Ramos, M.C., Bindi, M., Ginaldi, F., 2018. A model library to simulate grapevine growth and development: software implementation, sensitivity analysis and field level application. Eur. J. Agron. 99, 92–105. https://doi.org/10.1016/j.eja.2018.06.006.
Li-Mallet, A., Rabot, A., Geny, L., 2015. Factors controlling inflorescence primordia formation of grapevine: their role in latent bud fruitfulness? A review. Botany. 94, 147–163. https://doi.org/10.1139/cjb-2015-0108.
Lundell, R., Hänninen, H., Saarinen, T., Åström, H., Zhang, R., 2020. Beyond rest and quiescence (endodormancy and ecodormancy): A novel model for quantifying plant–environment interaction in bud dormancy release. Plant Cell Environ. 43, 40–54. https://doi.org/10.1111/pce.13650.
Magarey, R.D., Sutton, T.B., Thayer, C.L., 2005. A simple generic infection model for foliar fungal plant pathogens. Phytopathology. 95, 92–100. https://doi.org/10.1094/PHYTO-95-0092.
Magro, R.B., Alves, S.A.M., Gebler, L., 2023. Computational models in precision fruit growing: reviewing the impact of temporal variability on perennial crop yield assessment. SN Comput. Sci. 4, 554. https://doi.org/10.1007/s42979-023-02103-6.
Manzini, J., Hoshika, Y., Sicard, P., De Marco, A., Ferrini, F., Pallozzi, E., Neri, L., Baraldi, R., Paoletti, E., Baesso Moura, B., 2024. Detection of morphological and eco-physiological traits of ornamental woody species to assess their potential Net O3 uptake. Environ. Res. 252, 118844. https://doi.org/10.1016/j.envres.2024.118844.
Massano, L., Fosser, G., Gaetani, M., Bois, B., 2023. Assessment of climate impact on grape productivity: A new application for bioclimatic indices in Italy. Sci. Total Environ. 905, 167134. https://doi.org/10.1016/j.scitotenv.2023.167134.
Molitor, D., Fraga, H., Junk, J., 2020. UniPhen – a unified high resolution model approach to simulate the phenological development of a broad range of grape cultivars as well as a potential new bioclimatic indicator. Agric. For. Meteorol., 108024 https://doi.org/10.1016/j.agrformet.2020.108024.
Molitor, D., Keller, M., 2016. Yield of Müller-Thurgau and Riesling grapevines is altered by meteorological conditions in the current and previous growing seasons. OENO One 50, 4. https://doi.org/10.20870/oeno-one.2016.50.4.1071.
Moreira, G., dos Santos, F.N., Cunha, M., 2025. Grapevine inflorescence segmentation and flower estimation based on Computer Vision techniques for early yield assessment. Smart Agric. Technol. 10, 100690. https://doi.org/10.1016/j.atech.2024.100690.
Omazić, B., Prtenjak, M.T., Bubola, M., Meštrić, J., Karoglan, M., Prša, I., 2023. Application of statistical models in the detection of grapevine phenology changes. Agric. For. Meteorol. 341, 109682. https://doi.org/10.1016/j.agrformet.2023.109682.
Palacios, F., Diago, M.P., Melo-Pinto, P., Tardaguila, J., 2022. Early yield prediction in different grapevine varieties using computer vision and machine learning. Precis. Agric. 24, 407–435. https://doi.org/10.1007/s11119-022-09950-y.
Parker, A.K., García de Cortázar-Atauri, I., Trought, M.C., Destrac, A., Agnew, R., Sturman, A., van Leeuwen, C., 2020. Adaptation to climate change by determining grapevine cultivar differences using temperature-based phenology models. OENO One 54, 955–974. https://doi.org/10.20870/oeno-one.2020.54.4.3861.
Pei, J., Tan, S., Zou, Y., Liao, C., He, Y., Wang, J., Huang, H., Wang, T., Tian, H., Fang, H., Wang, L., Huang, J., 2025. The role of phenology in crop yield prediction: Comparison of ground-based phenology and remotely sensed phenology. Agric. For. Meteorol. 361, 110340. https://doi.org/10.1016/j.agrformet.2024.110340.
Petrie, P.R., Clingeleffer, P.R., 2005. Effects of temperature and light (before and after budburst) on inflorescence morphology and flower number of Chardonnay grapevines (Vitis vinifera L.). Aust. J. Grape Wine Res. 11, 59–65. https://doi.org/10.1111/j.1755-0238.2005.tb00279.x.
Raab, N., Meza, F.J., Franck, N., Bambach, N., 2015. Empirical stomatal conductance models reveal that the isohydric behavior of an Acacia caven Mediterranean Savannah scales from leaf to ecosystem. Agric. For. Meteorol. 213, 203–216. https://doi.org/10.1016/j.agrformet.2015.06.018.
Reis, S., Fraga, H., Carlos, C., Silvestre, J., Eiras-Dias, J., Rodrigues, P., Santos, J.A., 2020. Grapevine phenology in four Portuguese wine regions: modeling and predictions. Appl. Sci. 10, 3708. https://doi.org/10.3390/app10113708.
Roberts, A.M., Tansey, C., Smithers, R.J., Phillimore, A.B., 2015. Predicting a change in the order of spring phenology in temperate forests. Glob. Change Biol. 21, 2603–2611. https://doi.org/10.1111/gcb.12896.
Santos, J.A., Malheiro, A.C., Pinto, J.G., Jones, G.V., 2012. Macroclimate and viticultural zoning in Europe: observed trends and atmospheric forcing. Clim. Res. 51, 89–103. https://doi.org/10.3354/cr01056.
Santos, J.A., Ceglar, A., Toreti, A., Prodhomme, C., 2020. Performance of seasonal forecasts of Douro and Port wine production. Agric. For. Meteorol. 291, 108095. https://doi.org/10.1016/j.agrformet.2020.108095.
Simmonds, E.G., Cole, E.F., Sheldon, B.C., 2019. Cue identification in phenology: A case study of the predictive performance of current statistical tools. J. Anim. Ecol. 88, 1428–1440. https://doi.org/10.1111/1365-2656.13038.
Taylor, J.A., Tisseyre, B., Leroux, C., 2019. A simple index to determine if within-field spatial production variation exhibits potential management effects: Application in vineyards using yield monitor data. Precis. Agric. 20, 880–895. https://doi.org/10.1007/s11119-018-9620-3.
Valeriano, T.T.B., de Souza Rolim, G., Manici, L.M., Giustarini, L., Bregaglio, S., 2021. A process-based model to simulate sugarcane orange rust severity from weather data in Southern Brazil. Int. J. Biometeorol. 65, 2037–2051. https://doi.org/10.1007/s00484-021-02162-5.
van de Pol, M., Bailey, L.D., McLean, N., Rijsdijk, L., Lawson, C.R., Brouwer, L., 2016. Identifying the best climatic predictors in ecology and evolution. Methods Ecol. Evol. 7, 1246–1257. https://doi.org/10.1111/2041-210X.12590.
Vasconcelos, M.C., Greven, M., Winefield, C.S., Trought, M.C.T., Raw, V., 2009. The flowering process of Vitis vinifera: a review. Am. J. Enol. Vitic. 60, 411–434. https://doi.org/10.5344/ajev.2009.60.4.411.
Venables, W.N., Ripley, B.D., 2002. Modern Applied Statistics with S, Fourth edition. Springer. https://doi.org/10.1007/978-0-387-21706-2.
Victorino, G.F., Braga, R., Santos-Victor, J., Lopes, C.M., 2020. Yield components detection and image-based indicators for non-invasive grapevine yield prediction at different phenological phases. OENO One 54, 4. https://doi.org/10.20870/oeno-one.2020.54.4.3616.
Webber, H., Rezaei, E.E:, Ryo, M., Ewert, F., 2022. Framework to guide modeling single and multiple abiotic stresses in arable crops. Agric. Ecosyst. Environ. 340, 108179. https://doi.org/10.1016/j.agee.2022.108179.
Williams, L.E., Levin, A.D., Fidelibus, M.W., 2022. Crop coefficients (Kc) developed from canopy shaded area in California vineyards. Agric. Water Manag. 271, 107771. https://doi.org/10.1016/j.agwat.2022.107771.
Yan, W., Hunt, L.A., 1999. An equation for modelling the temperature response of plants using only the cardinal temperatures. Ann. Bot. 84, 607–614. https://doi.org/10.1006/anbo.1999.0955.

Zapata, D., Salazar-Gutierrez, M., Chaves, B., Keller, M., Hoogenboom, G., 2017. Predicting Key Phenological Stages for 17 Grapevine Cultivars (Vitis vinifera L.). Am. J. Enol. Vitic. 68, 60–72. https://doi.org/10.5344/ajev.2016.15077.

Zhu, J., Fraysse, R., Trought, M.C., Raw, V., Yang, L., Greven, M., Martin, D., Agnew, R., 2020. Quantifying the seasonal variations in grapevine yield components based on pre- and post-flowering weather conditions. OENO One 54. https://doi.org/10.20870/oeno-one.2020.54.2.2926.

Zhu, J., Parker, A., Gou, F., Agnew, R., Yang, L., Greven, M., Raw, V., Neal, S., Martin, D., Trought, M.C.T., Huth, N., Brown, H.E., 2021. Developing perennial fruit crop models in APSIM Next Generation using grapevine as an example. In. Silico Plants. 3, diab021. https://doi.org/10.1093/insilicoplants/diab021.